

Exercise 1: Self-induction and Spark-plug

- a) An electrical switch consists of two contacts that either touch each other or are separated by such a distance that no electrical current can flow as the space between is insulating. However there will be a break through if either the voltage is too high or the distance between the contacts is too small. In case the insulator is air nothing serious will happen, only the contacts will corrode with time. However, if an explosive gas is present it might be ignited by the spark. Therefore any closing of a switch is in such conditions dangerous, unless one closes the switch very fast and exactly at the time of a zero-crossing of an AC-voltage.

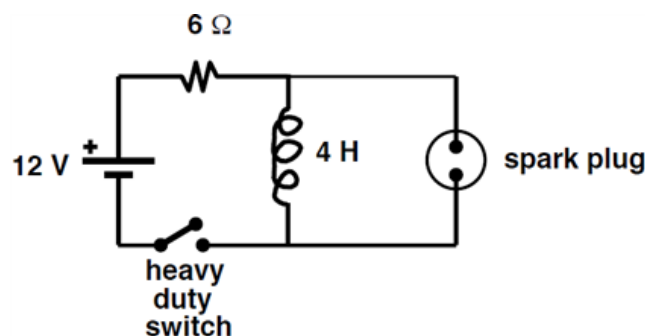
Since every current induces a magnetic field, any change of current leads to a change of magnetic field and to an inductance. Every electrical circuit thus has some degree of inductance attached to it even a simple straight wire. Upon a change in the flowing current, a voltage will be induced to compensate for the changing magnetic field. The relation which applies is the following:

$$u_{ind} = L \cdot \frac{dI}{dt} \quad (1)$$

where L is the inductance of the circuit. This voltage is now not only proportional to the inductance, but also the change of current over time, i.e. the faster a current is changed over time, the higher the voltage. Such induced voltages, depending on the circuit, can reach very large values (of the order of 1000 V). Thus the danger of having a spark as described above is significantly increased. In modern installations the danger might be reduced by a capacitance that is connected in parallel to the inductance.

Note that these induced voltages can not only affect a circuit itself (the case of self-inductance) but also nearby electrical circuits (mutual inductance).

- b) A spark-plug in a car makes use of exactly this self-inductance. On a very qualitative basis, the circuit of a spark-plug could look something like the one shown in the figure. The choice of the resistance and the inductance in this example then determines the voltage and thus the possibility of a spark.



Exercise 2: Resonance curve

a) The total impedance of the circuit is

$$Z_{\text{tot}} = Z_1 + \frac{1}{\frac{1}{Z_2} + \frac{1}{Z_3}} = Z_1 + \frac{Z_2 Z_3}{Z_2 + Z_3}. \quad (2)$$

Thus the current flowing through Z_1 is:

$$I_1 = \frac{V}{Z_{\text{tot}}} = \frac{V}{Z_1 + \frac{Z_2 Z_3}{Z_2 + Z_3}} = \frac{V(Z_2 + Z_3)}{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3}. \quad (3)$$

For the voltage drop over Z_3 we have

$$V_3 = Z_3 I_3 = V - Z_1 I_1 \quad (4)$$

and we obtain the current through Z_3 as:

$$I_3 = \frac{V - Z_1 I_1}{Z_3} = \frac{V - Z_1 \frac{V(Z_2 + Z_3)}{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3}}{Z_3} \quad (5)$$

$$= \frac{Z_2 V}{Z_1 Z_2 + Z_1 Z_3 + Z_2 Z_3} = \frac{V_0 e^{i\omega t}}{Z_1 + Z_3 + \frac{Z_1 Z_3}{Z_2}}, \quad (6)$$

where $V = V_0 e^{i\omega t}$ was put in.

b) Plugging in the Impedances from the right circuit into equation (6) yields:

$$I_3 = \frac{V_0 e^{i\omega t}}{\frac{1}{i\omega C} + R + \frac{R}{i\omega C \cdot i\omega L}} = \frac{\frac{\omega^2 V_0}{R} e^{i\omega t}}{\omega^2 - \frac{1}{LC} - i \frac{\omega}{RC}}. \quad (7)$$

Hence the phase difference between V and I_3 is given by

$$\tan \alpha = \frac{\text{Im} \left(\frac{\frac{\omega^2}{R}}{\omega^2 - \frac{1}{LC} - i \frac{\omega}{RC}} \right)}{\text{Re} \left(\frac{\frac{\omega^2}{R}}{\omega^2 - \frac{1}{LC} - i \frac{\omega}{RC}} \right)} = \frac{\frac{\omega}{RC}}{\omega^2 - \frac{1}{LC}} \quad (8)$$

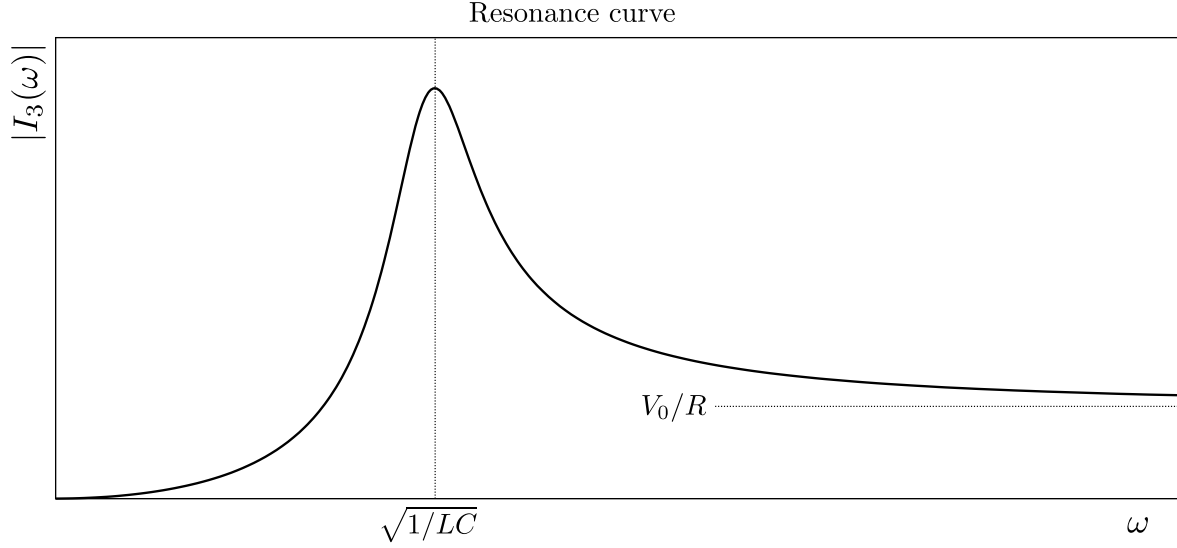
and the modulus of the current is

$$|I_3| = \frac{\frac{\omega^2 V_0}{R}}{\sqrt{\left(\omega^2 - \frac{1}{LC}\right)^2 + \left(\frac{\omega}{RC}\right)^2}}. \quad (9)$$

We analyse:

- for $\omega \rightarrow \infty$ we find $|I_3| \rightarrow V_0/R$, because for high frequencies the impedance of capacitor and coil approach zero and infinity, respectively.
- for $\omega = 0$ we find $|I_3| \rightarrow 0$, because for high frequencies the impedance of capacitor and coil approach infinity and zero, respectively.
- in case of a large resistor R we find a maximum for $|I_3|$ close to the frequency $\omega = 1/\sqrt{LC}$.

From these observation we can sketch $|I_3|$ as a function of ω .



Exercise 3: Disc in a magnetic field

- a) The flux through the disk is given by

$$\Phi = \int_A \vec{B} \cdot d\vec{A} \quad (10)$$

Because $d\vec{A}$ is in $\vec{e}_z$ -direction only the z -component of the magnetic field contributes to the flux.

$$\Phi = \int_A (k - 2z) B_0 t \, dA \quad (11)$$

where B_0 is a constant having units of $[B_0] = \text{T/ms}$.

Because the disc is placed at $z = 0$ we find

$$\Phi = \int_A k B_0 t \, dA \quad (12)$$

$$= \pi R^2 k B_0 t. \quad (13)$$

- b) We can use Faraday's law to determine the electric field at a certain radius r :

$$\oint \vec{E}_{ind} \cdot d\vec{l} = -\frac{d}{dt} \int \vec{B} \cdot d\vec{A} \quad (14)$$

We know that $\vec{E}_{ind}$ points in $\vec{e}_\phi$ direction, and thus, is always parallel to $d\vec{l}$.

$$\int_0^{2\pi} E_{ind}(r) r \, d\phi = -\frac{d}{dt} (k B_0 t r^2 \pi) \quad (15)$$

$$2\pi E_{ind}(r) r = -k B_0 r^2 \pi \quad (16)$$

$$E_{ind}(r) = -\frac{k B_0}{2} r \quad (17)$$

$$\vec{E}_{ind}(r) = -\frac{k B_0}{2} r \vec{e}_\phi \quad (\text{clockwise direction})$$

(18)

Alternative solution The following vectors are in cylinder coordinates.

$$\vec{\nabla} \times \vec{E} = -\frac{\partial}{\partial t} \vec{B} \quad (19)$$

$$\begin{pmatrix} \frac{1}{r} \frac{\partial E_z}{\partial \phi} - \frac{\partial E_\phi}{\partial z} \\ \frac{\partial E_r}{\partial z} - \frac{\partial E_z}{\partial r} \\ \frac{1}{r} \left(\frac{\partial(rE_\phi)}{\partial r} - \frac{\partial E_r}{\partial \phi} \right) \end{pmatrix} = \begin{pmatrix} -rB_0 \\ 0 \\ -(k-2z)B_0 \end{pmatrix} \quad (20)$$

The cancellation of $\partial_\phi E_z$ to 0 occurs because we know from the given $\vec{E}$ field that inside the disk (at $z = 0$) the z component has the value 0 for any ϕ and hence no dependence on ϕ . The same is true for $\partial_r E_z$ and $\partial_\phi E_r$. We cannot make any statement about z -dependencies, as we know the $\vec{E}$ field only at $z = 0$.

From the $\vec{e}_r$ component we get the equation

$$\frac{\partial E_\phi}{\partial z} = rB_0 \quad \left| \cdot \partial z \right| \int \quad (21)$$

$$E_\phi = rB_0 z + m \quad (22)$$

where m is a constant which cannot depend on z but is otherwise unknown. This is not really helpful yet.

From the $\vec{e}_z$ component we get the equation

$$\frac{1}{r} \frac{\partial(rE_\phi)}{\partial r} = -(k-2z)B_0 \quad \left| \cdot r \right| \left| \cdot \partial r \right| \int \quad (23)$$

$$rE_\phi = -(k-2z)\frac{1}{2}r^2B_0 + l \quad (24)$$

$$E_\phi = -k\frac{1}{2}rB_0 + rB_0z + l/r \quad (25)$$

where l is a constant which cannot depend on r but is otherwise unknown. Comparing with (22) that $m = -kB_0r/2 + l$ where l must be independent from r and z . As we can request rotational symmetry, it must be also independent from ϕ .

Side note: The undetermined term l/r arises from a division 0 by 0 in the differential equation in cylindrical coordinates. Calculating this in detail exceeds the skills of a first-year student.

In order to have a finite E-field at $r = 0$ we have to choose $l = 0$, which leads to the solution:

$$E_\phi = -\frac{B_0}{2}r(k-2z) \quad (26)$$

And thus for $z = 0$ we obtain

$$E_\phi = -\frac{B_0}{2}rk \quad (27)$$

c) The potential difference between $r = 0$ and $r = R$ is:

$$\Delta V = \int_0^R \vec{E} \cdot d\vec{r} = \int_0^R -\frac{kB_0}{2}r dr \vec{e}_\phi \cdot \vec{e}_r = \boxed{0} \quad (28)$$

d) The force on the ring with radius r is

$$dF = dq \cdot E_{ind}(r), \quad (29)$$

where

$$dq = 2\pi r dr \cdot \underbrace{\frac{Q}{R^2\pi}}_{\text{surface density}} \quad (30)$$

is the charge on a ring of radius r and thickness dr , so

$$dF = \frac{Q}{R^2\pi} 2\pi r dr \cdot \frac{-kB_0}{2} r = -kB_0 \frac{Q}{R^2} r^2 dr. \quad (31)$$

($d\vec{F}$ pulls to the $-\phi$ -direction).

The torque $d\vec{\tau} = \vec{r} \times d\vec{F}$ acting on this ring is pointing parallel to the z -direction and has a value of:

$$d\tau = dF r = -kB_0 \frac{Q}{R^2} r^3 dr. \quad (32)$$

The total torque is just the integral of $d\tau$ from $r = 0$ to $r = R$:

$$\boxed{\tau = -kB_0 \frac{Q}{R^2} \int_0^R r^3 dr = -\frac{kB_0 Q}{4} R^2.} \quad (33)$$

The minus sign tells us that the torque points in $-z$ -direction.

e) The angular momentum of the disc at time t is in z -direction and is given by

$$L(t) = \int_0^t \tau(t') dt' = -\frac{kB_0 Q}{4} R^2 t. \quad (34)$$

The angular velocity is

$$\boxed{\omega(t) = \frac{L(t)}{\frac{1}{2}mR^2} = -\frac{kB_0 Q}{2m} t.} \quad (35)$$

f) The magnetic field produced by one ring with radius r is

$$dB_z = \frac{\mu_0}{2} \frac{dI}{r}, \quad (36)$$

where dI is the current in the ring:

$$dI = dq \cdot \frac{\omega}{2\pi} \quad (37)$$

$$= 2\pi r dr \frac{Q}{R^2\pi} \cdot \frac{-kB_0 Q}{2m} t \cdot \frac{1}{2\pi} \quad (38)$$

$$= -\frac{kB_0}{2\pi} \frac{Q^2}{mR^2} t r dr. \quad (39)$$

Therefore the magnetic field produced by a ring of width dr is

$$dB_z = -\frac{\mu_0 k B_0}{4\pi} \frac{Q^2}{mR^2} t dr. \quad (40)$$

The total magnetic field is then:

$$B_z = \int_0^R dB_z = -\frac{\mu_0 k B_0}{4\pi} \frac{Q^2}{mR} t. \quad (41)$$

Exercise 4: Maxwell's displacement current

With the help of the tip, one can calculate the rotation of the magnetic field, as its only component is in ϕ -direction:

$$\begin{aligned} \vec{\nabla} \times \vec{B} &= \left(-\frac{\partial B_\phi}{\partial z} \right) \hat{r} + \frac{1}{r} \left(\frac{\partial(rB_\phi)}{\partial r} \right) \hat{z} = \\ &= \frac{1}{r} \left[\frac{\partial \left((r\mu_0 V / 2\pi R) e^{-t/RC} \left(\frac{r}{r_0^2} \right) \right)}{\partial r} \right] \hat{z} = \\ &= \frac{1}{r} \left[2r \frac{\mu_0 V}{2\pi R} e^{-t/RC} \frac{1}{r_0^2} \right] \hat{z} = \\ &= \left[\frac{\mu_0 V}{\pi R} e^{-t/RC} \frac{1}{r_0^2} \right] \hat{z}. \end{aligned}$$

Since there is no other current between the plates ($\vec{J} = 0$), the Maxwell equation reduces to

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\epsilon_0 \frac{\partial \vec{E}}{\partial t} \right),$$

out of which one can deduce the displacement current,

$$\epsilon_0 \frac{\partial \vec{E}}{\partial t} = \frac{\vec{\nabla} \times \vec{B}}{\mu_0} = \frac{1}{\mu_0} \left[\frac{\mu_0 V}{\pi R} e^{-t/RC} \frac{1}{r_0^2} \right] \hat{z} = \left[\frac{V}{R} e^{-t/RC} \frac{1}{\pi r_0^2} \right] \hat{z}.$$

It can be seen, that the displacement current flows in z-direction.

Exercise 5: Maxwell's equations and EM waves

a) From $\vec{\nabla} \cdot \vec{B} = 0$ one obtains $B_{0x} = 0$. The partial derivative by time of $\vec{B}$ is

$$\vec{B} = \begin{pmatrix} 0 \\ \omega B_{0y} \sin(kx - \omega t + \phi_y) \\ \omega B_{0z} \sin(kx - \omega t + \phi_z) \end{pmatrix}$$

and with $\vec{\nabla} \times \vec{E} = -kE_0 \sin(kx - \omega t) \vec{e}_z$ we find $B_{0y} = 0$, $-kE_0 = \omega B_{0z}$ and $\phi_z = 0$. Thus, $\vec{B} = \frac{k}{\omega} E_0 \cos(kx - \omega t) \vec{e}_z$.

b) From $\vec{\nabla} \times \vec{B} = \frac{k^2}{\omega} E_0 \sin(kx - \omega t) \vec{e}_y$ and $\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} = \omega \mu_0 \epsilon_0 E_0 \sin(kx - \omega t) \vec{e}_y$ it follows that

$$\frac{\omega}{k} = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}.$$

This can be used to calculate the speed of light by measuring the electromagnetic force of a given configuration of charges and currents in a laboratory.

Maxwell himself did this 1870 and he was the first who brought this measurement into the context of a measurement of the speed of light. He used charges, batteries, coils and currents and identified light as a form of electromagnetic waves. It is also interesting to know that according to international conventions, ϵ_0 is defined via μ_0 and c .

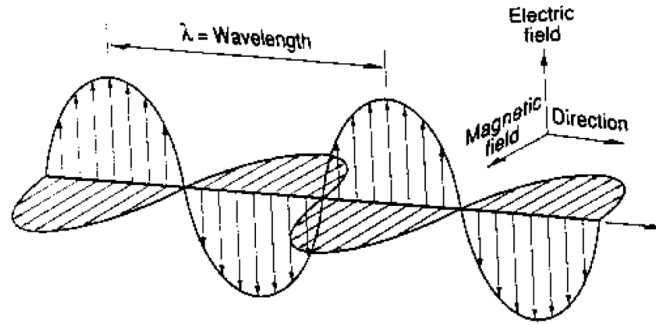


Abbildung 5.1: electromagnetic wave

c)

d) for $\phi = 90^\circ$, the total field is $\vec{E}^{\text{tot}} = \vec{E} + \vec{E}' = E_0 \cos(kx - \omega t) \vec{e}_y + E_0 \sin(kx - \omega t) \vec{e}_z$. The angle θ between the y -axis and $\vec{E}$, suffices the following condition:

$$\tan \theta = \frac{E_z^{\text{tot}}}{E_y^{\text{tot}}} = \tan(kx - \omega t). \quad (42)$$

Therefore, $\theta = \text{const} - \omega t$. This equates to a circularly polarized wave. Similarly, it follows for a phase $\phi = 0$ that the angle between the y -axis and $\vec{E}$, which is $\theta = 45^\circ$, which in turn corresponds to a linearly polarized wave.